A Quality-Controlled Semi-Supervised Detection Framework
for Cross-Domain Oil Palm
Fresh Fruit Bunch Maturity Grading

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Abstract

Automated oil palm fresh fruit bunch (FFB) maturity grading can support harvesting decisions, reduce manual subjectivity, and improve traceability in plantation operations. However, visual detection models trained on one image source often degrade when deployed across plantations, acquisition devices, lighting conditions, and dataset annotation protocols. This study presents a quality-controlled semi-supervised object detection framework for cross-domain oil palm FFB maturity grading. The framework starts from a YOLOv8n detector trained on a labeled source-domain video-frame dataset, introduces class-space harmonization across heterogeneous datasets, applies scene-disjoint evaluation to reduce frame-level leakage, and evaluates external-domain performance on a separate preprocessed FFB dataset with a four-class masked protocol. To improve robustness, the proposed pipeline combines class-balanced target-domain adaptation with strict pseudo-label filtering based on confidence, bounding-box geometry, edge-contact penalties, folder-level weak-label consistency, multi-view augmentation consistency, and class-wise dynamic confidence thresholds. Three random seeds were used for the main formal comparison. The source-only baseline achieved high scene-disjoint source-domain performance but collapsed on the locked external domain, with external mAP@0.5 of 0.0372 and mAP@0.5:0.95 of 0.0130, confirming severe domain shift. A class-balanced target-domain adaptation model improved external mAP@0.5 to 0.6973 and mAP@0.5:0.95 to 0.4758. A precision-first strict SSOD model achieved calibrated external mAP@0.5 of 0.7240, mAP@0.5:0.95 of 0.4863, precision of 0.8753, recall of 0.5364, and F1-score of 0.6321. A third-domain zero-shot image-level evaluation further indicated improved transfer relative to the source-only baseline. The results show that external-domain adaptation and quality-controlled pseudo-label selection can produce a more conservative and field-oriented FFB maturity detector, although under-ripe fruit bunches remain the main error source.

Keywords: agricultural engineering; domain adaptation; fresh fruit bunch; maturity grading; object detection; oil palm

1 Introduction

Oil palm fresh fruit bunch (FFB) maturity grading is a key operational step in plantation management and palm oil production. Harvesting unripe bunches reduces oil extraction efficiency, whereas delayed harvesting can increase overripe bunches, loose fruit loss, and quality degradation. In practice, FFB maturity is often judged manually from color, texture, and fruit detachment cues. Manual grading is inexpensive and flexible, but it is also subjective, labor-intensive, and sensitive to worker experience, illumination, occlusion, and the mixture of bunches in piles or field scenes (Suharjito et al., 2023; Omar et al., 2024).

Computer vision has therefore been widely explored for agricultural quality assessment, fruit detection, and maturity grading (Bargoti and Underwood, 2017; Kamilaris and Prenafeta-Boldu, 2018). Modern object detectors, especially one-stage detectors in the YOLO family, are attractive for field applications because they provide a practical compromise between accuracy and inference speed (Redmon et al., 2016; Jocher et al., 2023). For oil palm FFB maturity detection, a detector must localize bunches and distinguish visually similar maturity categories such as unripe, under-ripe, ripe, and overripe. Previous oil palm studies have shown the feasibility of deep learning and YOLO-based approaches for FFB ripeness classification or detection (Suharjito et al., 2021; Lai et al., 2022; Mansour et al., 2022). However, this task remains more difficult than simple fruit presence detection because the decision boundary depends on fine color transitions, fruit surface texture, shadows, and camera exposure.

A major limitation of many agricultural detection studies is that the reported performance is often measured on random splits or same-source test sets. For video-frame datasets, random splitting can overestimate performance because adjacent frames from the same video, pile, lighting condition, or acquisition session may appear in both training and testing subsets. A model may therefore learn scene-specific appearance rather than robust maturity cues. When deployed on another dataset or plantation scene, performance can drop sharply. This issue is especially important for oil palm FFB maturity grading because available datasets are heterogeneous: some are collected from videos of fruit bunch piles (Suharjito et al., 2023), others from outdoor plantation scenes (Omar et al., 2024), and others are preprocessed into class folders or weakly labeled subsets.

This study reframes oil palm FFB maturity detection as a cross-domain agricultural engineering problem rather than a single-dataset YOLO training task. The central hypothesis is that a source-domain detector trained on video-frame data can perform well in-domain but fail on an external domain, and that this failure can be mitigated by combining target-domain adaptation with quality-controlled semi-supervised object detection (SSOD). The proposed framework is designed to be conservative: pseudo-labels are not accepted solely from high detector confidence, but must also satisfy geometry constraints, edge-contact penalties, weak class-folder consistency, multi-view augmentation consistency, and class-wise dynamic thresholds.

The main contributions of this work are as follows. First, a strict evaluation protocol is introduced for oil palm FFB maturity detection, including source-domain scene-disjoint evaluation, locked external-domain testing, and four-class masked evaluation when the target dataset lacks empty and overripe categories. Second, a class-space harmonization strategy is used to integrate source and external datasets with inconsistent maturity labels while avoiding invalid class penalties in missing-class datasets. Third, a quality-controlled SSOD pipeline is implemented to filter pseudo-labeled images under multi-view consistency and class-balanced selection. Fourth, formal three-seed experiments are reported for a source-only baseline, a class-balanced target-domain adaptation baseline, and a precision-first strict SSOD model. The results are interpreted conservatively: the large gain from source-only to target-adapted models is evidence of domain adaptation, not zero-shot generalization.

2 Materials and Methods

2.1 Datasets and Class Harmonization

The experimental pipeline used three dataset roles, summarized in Table 1. The first was a labeled source-domain oil palm FFB dataset based mainly on ScienceDB/OILPALM-style video-frame imagery (Suharjito et al., 2023). This dataset was used to train the source-only supervised baseline and to construct a scene-disjoint split. The second was an external-domain preprocessed FFB dataset, which was used as the main target-domain dataset for cross-domain evaluation and adaptation. The third was an outdoor oil palm ripeness dataset used only for exploratory third-domain zero-shot image-level evaluation (Omar et al., 2024).

The unified class scheme contained six maturity or condition classes (Table 2):

$$\mathcal{C} = \{\text{abnormal}, \text{empty}, \text{overripe}, \text{ripe}, \text{under\_ripe}, \text{unripe}\}.$$

Because the external preprocessed FFB dataset did not contain empty and overripe classes, the external evaluation was performed under a masked four-class protocol:

$$\mathcal{C}_{ext} = \{\text{abnormal}, \text{ripe}, \text{under\_ripe}, \text{unripe}\}.$$

Empty and overripe predictions were ignored for this external protocol. This prevents the model from being penalized for classes that were absent from the external dataset and allows a fairer comparison across heterogeneous label spaces.

To reduce overly optimistic evaluation caused by adjacent video frames, the source-domain data were not used only under a random split. A video- or scene-disjoint split was constructed based on filename prefixes or inferred video identifiers. Frames from the same scene group were assigned to the same split. The official or original source-domain split was retained only as an in-domain reference and was not used as the main conclusion.

2.2 Model Stages

Four model stages were considered during development, but the formal manuscript focuses on three main stages.

Source-only YOLOv8n baseline. The first model was a YOLOv8n detector initialized from COCO-pretrained weights and trained only on the source-domain labeled dataset. This model represents the common supervised baseline and is used to quantify the cross-domain collapse when evaluated on the external FFB dataset.

Class-balanced target-domain adaptation model. The second model, denoted Model2 Balanced, used source-domain data together with a class-balanced supervised subset of the external-domain training data. This model is not a zero-shot model; it represents a practical target-domain adaptation baseline. It was included because comparing the final system only against the weak source-only baseline would exaggerate the improvement.

Precision-first strict SSOD model. The third model applied quality-controlled pseudo-label self-training on top of the adapted teacher. It was designed for high-precision deployment, where false positives in maturity grading are operationally costly. The SSOD stage used multi-view consistency, quality scoring, dynamic class-wise thresholds, and class-balanced pseudo-label acceptance.

2.3 Pseudo-Label Generation with Multi-View Consistency

For each candidate unlabeled or weakly labeled image, the teacher detector generated predictions on the original image and on augmented views. Augmentations included brightness and contrast jitter, horizontal or vertical flips, small rotations, and crop-like perturbations. Predictions from augmented views were mapped back to the original image coordinate system. A pseudo-label candidate was accepted only when boxes from the original and augmented views agreed in class and localization. The localization consistency was measured using intersection-over-union (IoU), and inconsistent predictions were rejected.

This design follows the general principle of consistency regularization in semi-supervised learning: a reliable object prediction should remain stable under label-preserving image transformations (Tarvainen and Valpola, 2017; Sohn et al., 2020a). In this work, consistency was used as a filtering criterion rather than a fully online teacher-student loss, making the method easier to reproduce with standard YOLOv8 training tools. The approach is related to pseudo-label and consistency-based SSOD methods (Sohn et al., 2020b; Liu et al., 2021; Xu et al., 2021; Li et al., 2022), but it is implemented as an offline quality-control layer for an agricultural engineering workflow.

2.4 Pseudo-Label Quality Score

Each predicted bounding box was assigned a composite quality score:

$$Q = w_c Q_c + w_g Q_g + w_e Q_e + w_f Q_f + w_m Q_m,$$

where Q_c is detector confidence, Q_g is a geometry score, Q_e is an edge-contact score, Q_f is weak folder-level class consistency, Q_m is multi-view consistency, and $w_c, \ldots, w_m$ are non-negative weights. The geometry score penalized boxes with implausible normalized area or aspect ratio. The edge score penalized boxes touching image boundaries, because such boxes often represent incomplete bunches or truncated detections. The folder-consistency score used class-folder information only as an image-level weak prior, not as bounding-box ground truth. This distinction is important: the method is weak-label-guided SSOD rather than purely unlabeled SSOD.

Pseudo-labels were filtered by both a global quality threshold and class-wise confidence thresholds. Images were then selected under a class-balanced quota so that abundant easy classes did not dominate the pseudo-labeled training set.

2.5 Dynamic Class-Wise Thresholds

Class-wise thresholds were updated using validation statistics. Classes with weaker validation performance or higher confusion risk were assigned stricter confidence thresholds, while easier classes could use lower thresholds. This mechanism was intended to reduce noisy pseudo-labels for visually difficult categories, especially under-ripe and unripe bunches, which often share color and texture cues.

For deployment-style reporting, class-wise calibrated thresholds were also selected for a high-precision operating point. This is appropriate for maturity grading scenarios where a conservative detector may be preferred over a detector that maximizes recall at the cost of many false positives.

2.6 Training and Evaluation Protocol

All YOLOv8 training was performed using the Ultralytics Python API with Windows-compatible settings, including `workers=0`. The input resolution for formal external evaluation was 640 pixels. Three random seeds were used for the formal comparison: 42, 2026, and 3407.

The following evaluation layers were used:

1. **Scene-disjoint source-domain test**: evaluates source-domain generalization across held-out video or scene groups.

2. **Locked external-domain test**: evaluates cross-domain detection on the preprocessed FFB dataset using the four-class masked protocol.

3. **Calibration evaluation**: reports a high-precision external operating point with calibrated class-wise thresholds.

4. **Third-domain zero-shot evaluation**: evaluates image-level class prediction on an outdoor oil palm dataset that was not used for training.

The primary detection metrics were precision, recall, F1-score, mAP@0.5, and mAP@0.5:0.95. The third-domain experiment was reported separately as image-level top-1 accuracy and macro F1-score, and should not be compared directly with detection mAP.

3 Results

3.1 Domain Shift from Source-Only Training

The source-only YOLOv8n model achieved scene-disjoint source-domain mAP@0.5 of 0.5917 and mAP@0.5:0.95 of 0.5018 across three seeds (Table 3). However, when evaluated on the locked external preprocessed FFB dataset, performance dropped to mAP@0.5 of 0.0372 and mAP@0.5:0.95 of 0.0130. This confirms severe cross-domain degradation. Therefore, the source-only result was used as evidence of domain shift rather than as the main baseline for claiming method improvement.

The class-balanced target-domain adaptation model substantially improved external-domain performance, reaching external mAP@0.5 of 0.6973 and mAP@0.5:0.95 of 0.4758. This result demonstrates that using the external-domain training portion for adaptation can recover target-domain detection performance. It does not demonstrate zero-shot generalization, because external-domain data were used during training.

3.2 Strict SSOD and High-Precision Calibration

The precision-first strict SSOD model achieved scene-disjoint mAP@0.5 of 0.6373 and mAP@0.5:0.95 of 0.5408. Under calibrated external-domain evaluation, it achieved mAP@0.5 of 0.7240, mAP@0.5:0.95 of 0.4863, precision of 0.8753, recall of 0.5364, and F1-score of 0.6321 (Table 4; Fig. 1). Compared with the class-balanced adaptation baseline, strict SSOD mainly improved the high-precision operating point rather than producing a large headline mAP improvement. The calibration curve in Fig. 2 illustrates this precision-recall trade-off.

This behavior is consistent with the design of the method. Strict pseudo-label filtering and high class-wise thresholds reject uncertain detections, reducing false positives but also reducing recall. For field deployment, this may be desirable when the cost of incorrect maturity grading is high. For exhaustive counting or harvest-yield estimation, the lower recall would require further optimization.

3.3 Third-Domain Zero-Shot Image-Level Evaluation

On the outdoor third-domain dataset, the source-only model achieved image-level top-1 accuracy of 0.4632 and macro F1-score of 0.3480 (Table 5). Model2 Balanced improved top-1 accuracy to 0.6000 and macro F1-score to 0.4383. Strict SSOD further improved top-1 accuracy to 0.6351 but had lower macro F1-score of 0.4048. This indicates that strict SSOD improved overall image-level correctness but did not uniformly improve all classes. The third-domain dataset was relatively small, with 95 images per seed in the evaluation, so these results should be treated as exploratory evidence rather than definitive proof of generalization.

3.4 Under-Ripe Error Analysis

Under-ripe FFB remained the most difficult category. In the seed-2026 locked external analysis, Model2 Balanced achieved under-ripe AP@0.5 of 0.7528 and AP@0.5:0.95 of 0.5352, with 24 false-negative under-ripe cases and 8 misclassified under-ripe cases

(Table 6). Strict SSOD achieved lower under-ripe AP@0.5 of 0.6657 and AP@0.5:0.95 of 0.4918, with 48 false negatives and 5 misclassifications. This suggests that strict filtering reduced some class confusion but also made the detector more conservative for under-ripe bunches. The confusion matrices in Fig. 3 and Fig. 4 show the corresponding external-domain class-level behaviour.

The under-ripe limitation is technically plausible. Under-ripe bunches lie between unripe and ripe in the maturity continuum and can be visually ambiguous under different lighting conditions. Future improvements should focus on ordinal maturity modeling, targeted under-ripe positive sampling, and richer color-calibrated field data rather than only increasing the amount of pseudo-labeled data.

4 Discussion

The results support three engineering interpretations. First, random or same-source evaluation is insufficient for oil palm FFB maturity detection. The large gap between source-domain scene-disjoint performance and locked external-domain performance shows that a detector trained on one data source may not generalize to another without adaptation. This is especially important for video-frame datasets, where adjacent frames can produce optimistic random-split metrics.

Second, target-domain adaptation is the dominant source of external-domain recovery. The improvement from source-only external mAP@0.5 of 0.0372 to Model2 Balanced external mAP@0.5 of 0.6973 should not be presented as a pure SSOD gain. It reflects the transition from source-only training to target-domain adaptation using the external training portion. This interpretation is important for scientific validity and for avoiding inflated claims.

Third, strict SSOD contributes mainly to conservative deployment behavior. The precision-first model achieved high calibrated precision of 0.8753, but recall was 0.5364. This trade-off is relevant in agricultural engineering practice. If the system is used to flag high-confidence maturity classes for quality control, high precision is valuable. If the system is used to count all bunches or automate complete grading, additional recall-focused improvements are necessary.

From a practical workflow perspective, the proposed system is therefore better suited to human-in-the-loop decision support than to fully autonomous grading. A camera installed near a collection point, grading line, or mobile inspection station could use the calibrated detector to flag high-confidence abnormal, ripe, under-ripe, or unripe bunches for operator review. Low-confidence or ambiguous cases, especially under-ripe bunches near the ripe/unripe boundary, should remain subject to manual confirmation. This deployment mode matches the observed precision-recall trade-off and reduces the risk of presenting the system as a complete replacement for trained graders.

The proposed framework has several limitations. The external preprocessed FFB dataset lacks empty and overripe categories, requiring masked evaluation. Although this is methodologically necessary, it means the reported external-domain result does not cover the full six-class maturity scheme. The third-domain outdoor dataset was used only for image-level evaluation and had limited class balance, so it cannot replace a large object-detection external test set. Finally, the SSOD stage used offline pseudo-label filtering rather than a fully online exponential moving average teacher-student training loop. This choice improves reproducibility and engineering simplicity but may limit the maximum performance gain.

5 Conclusions

This study developed a quality-controlled semi-supervised YOLOv8 framework for cross-domain oil palm FFB maturity detection. The framework combines class-space harmonization, scene-disjoint evaluation, locked external-domain testing, class-balanced target-domain adaptation, multi-view pseudo-label consistency, composite pseudo-label quality scoring, dynamic class-wise thresholds, and high-precision calibration.

The experiments show that source-only training on video-frame data can fail severely on an external FFB dataset, confirming strong domain shift. Class-balanced target-domain adaptation substantially improves external detection performance. Strict SSOD provides a more conservative high-precision operating point, achieving calibrated external precision of 0.8753 with mAP@0.5 of 0.7240, although recall remains limited. Under-ripe detection remains the main bottleneck because it is visually close to neighboring maturity stages.

For practical deployment, the proposed framework is most suitable as a conservative decision-support system for FFB maturity grading rather than a complete replacement for human inspection. Future work should expand third-domain detection data, add ordinal maturity constraints, collect more under-ripe boundary cases, and evaluate the system on continuous field video streams.

Data Availability Statement

The source and external datasets used in this study were derived from publicly available or locally prepared oil palm FFB datasets. The manuscript repository contains the class-space definition, selected formal evaluation outputs, and the code needed to reproduce the preparation and evaluation protocol. Raw images and trained weights are not redistributed in the repository because their redistribution depends on the licenses and access conditions of the original datasets. Processed split manifests and additional evaluation outputs can be shared by the corresponding author upon reasonable request where permitted by the original data licenses.

Code Availability Statement

The implementation was developed in Python using the Ultralytics YOLOv8 API. The pipeline includes dataset preparation, scene-disjoint splitting, duplicate checking, pseudo-label generation, quality scoring, class-wise threshold calibration, and structured JSON/CSV reporting. The public repository is available at `https://github.com/yuningwuyn-lgtm/oil-palm-ffb-ssod`. The repository contains source code, configuration templates, manuscript files, and selected evaluation reports; raw third-party datasets and trained weights are excluded because their redistribution depends on the licenses and access conditions of the original data providers.

Conflict of Interest

The author declares no conflict of interest.

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CRediT Authorship Contribution

WU YUNING: conceptualization, methodology, software, validation, formal analysis, data curation, writing-original draft preparation, visualization, and project administration.

Declaration of Generative Artificial Intelligence and Artificial Intelligence-Assisted Technologies in the Writing Process

During the preparation of this work, the author used OpenAI ChatGPT/Codex to assist with language polishing, code organization, validation-checklist preparation, and manuscript formatting. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Supporting Agencies

No external supporting agency was involved in this study.

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Tables

Table 1: Dataset roles used in the cross-domain evaluation protocol.

Dataset role	Main use	Supervision type	Evaluation note
Source-domain video-frame FFB data	Source training and scene-disjoint testing	Bounding-box labels	Quantifies in-domain and video/scene generalization
External preprocessed FFB	Target-domain adaptation and locked testing	Four-class detection protocol	Empty and overripe are ignored in masked evaluation
Outdoor oil palm ripeness data	Third-domain exploratory testing	Image-level class labels	Used only for zero-shot image-level transfer analysis
Unlabeled or weakly labeled images	Pseudo-label self-training	Folder-level weak prior or unlabeled	Filtered by quality score and multi-view consistency

Table 2: Unified class scheme and external-domain masked evaluation setting.

Class ID	Unified class	Used in external masked evaluation
0	abnormal	Yes
1	empty	Ignored
2	overripe	Ignored
3	ripe	Yes
4	under_ripe	Yes
5	unripe	Yes

Table 3: Formal three-seed detection results. Values are mean ± standard deviation.

Model	Scene mAP50	Scene mAP50–95	Ext. mAP50	Ext. mAP50–95
Source-only YOLOv8n	0.5917 ± 0.0339	0.5018 ± 0.0343	0.0372 ± 0.0221	0.0130 ± 0.0075
Model2 Balanced	0.6459 ± 0.0190	0.5484 ± 0.0149	0.6973 ± 0.0298	0.4758 ± 0.0175
Stage-1 SSOD teacher	0.6141 ± 0.0340	0.5125 ± 0.0183	0.6619 ± 0.0283	0.4462 ± 0.0309
Stage-2 refined	0.5878 ± 0.0436	0.4900 ± 0.0395	0.6754 ± 0.0430	0.4645 ± 0.0331

Model	Ext. precision	Ext. recall	Ext. F1
Source-only YOLOv8n	0.1640 ± 0.0747	0.0933 ± 0.0530	0.0735 ± 0.0308
Model2 Balanced	0.6126 ± 0.0323	0.7527 ± 0.0121	0.6442 ± 0.0332
Stage-1 SSOD teacher	0.6163 ± 0.0468	0.7411 ± 0.0473	0.6390 ± 0.0174
Stage-2 refined	0.6294 ± 0.0402	0.7427 ± 0.0338	0.6743 ± 0.0386

Table 4: Precision-first strict SSOD calibrated external-domain results across three seeds.

Metric	Mean	Standard deviation
Scene-disjoint mAP@0.5	0.6373	0.0286
Scene-disjoint mAP@0.5:0.95	0.5408	0.0276
Calibrated external mAP@0.5	0.7240	0.0072
Calibrated external mAP@0.5:0.95	0.4863	0.0038
Calibrated external precision	0.8753	0.0116
Calibrated external recall	0.5364	0.0232
Calibrated external F1-score	0.6321	0.0294

Table 5: Third-domain zero-shot image-level evaluation. Detection mAP is not reported for this dataset because the protocol is image-level.

Model	Top-1 accuracy	Macro precision	Macro recall	Macro F1
Source-only YOLOv8n	0.4632 ± 0.0459	0.3936 ± 0.0690	0.4078 ± 0.0974	0.3480 ± 0.0565
Model2 Balanced	0.6000 ± 0.0380	0.4804 ± 0.1305	0.4585 ± 0.0905	0.4383 ± 0.1035
Strict SSOD	0.6351 ± 0.0338	0.4015 ± 0.0436	0.4451 ± 0.0275	0.4048 ± 0.0336

Table 6: Seed-2026 under-ripe class analysis on the locked external dataset.

Model	Under-ripe AP@0.5	Under-ripe AP@0.5:0.95	False negatives	Misclassified cases
Model2 Balanced	0.7528	0.5352	24	8
Strict SSOD	0.6657	0.4918	48	5

Figures

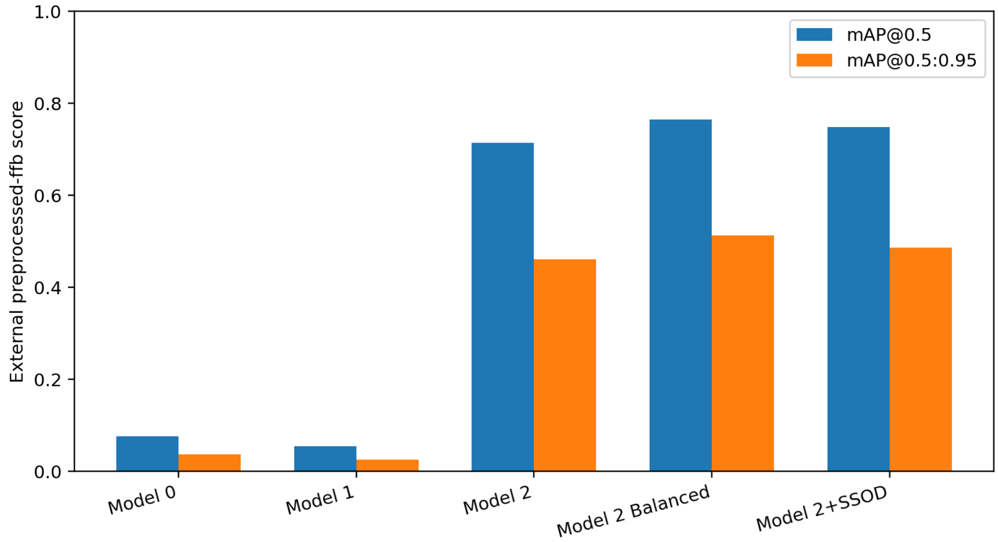

Figure 1: External-domain model comparison plot generated from the project reporting pipeline.

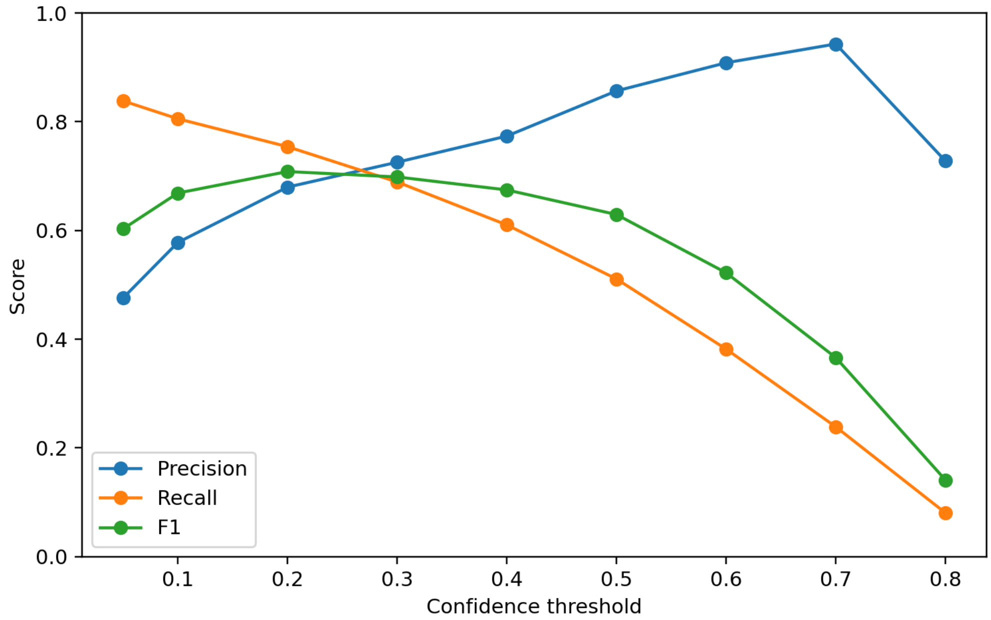

Figure 2: External-domain calibration curve used to select high-precision operating points.

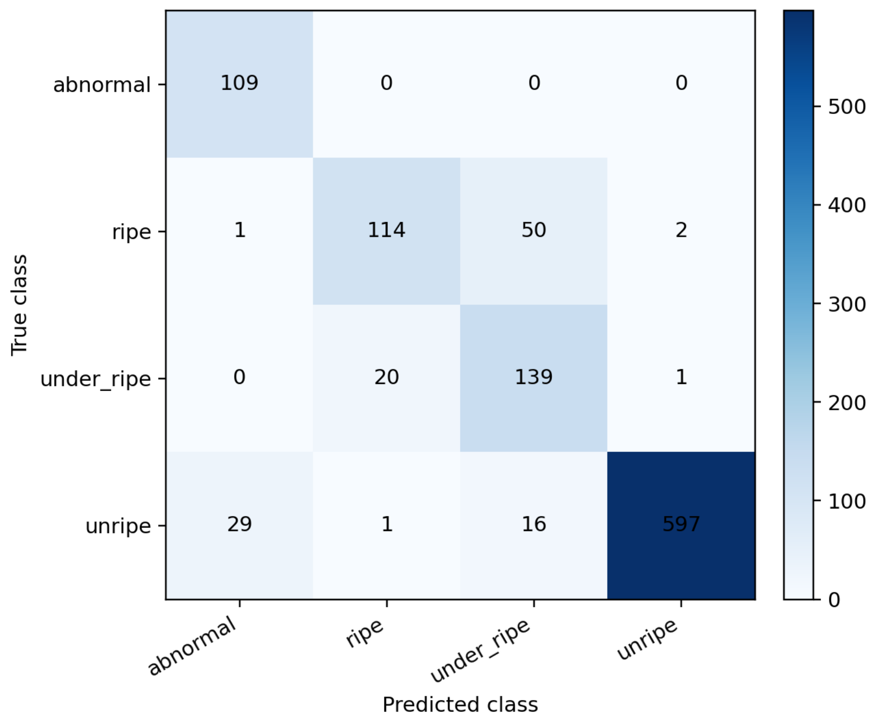

Figure 3: External-domain confusion matrix for the class-balanced target-domain adaptation model.

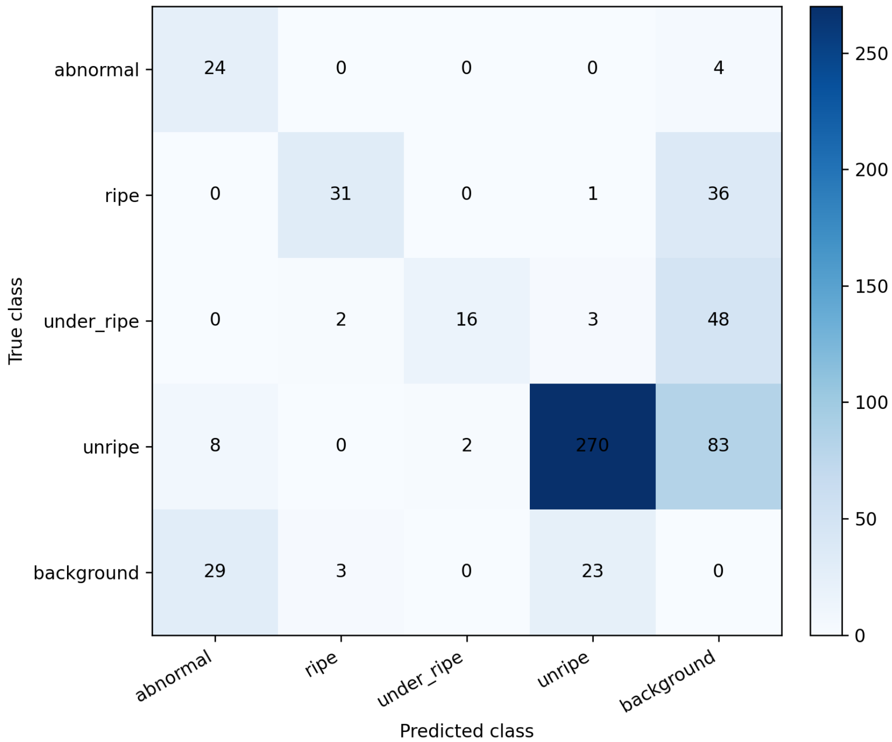

Figure 4: Locked external-domain confusion matrix for the strict SSOD model under the four-class masked protocol.